PERT CHARTS

PROJECT EVALUATION & REVIEW TECHNIQUE

Objectives

1. Learn how to Install **EdrawMax**.
2. Understand how to draw PERT Charts.

PERT Charts

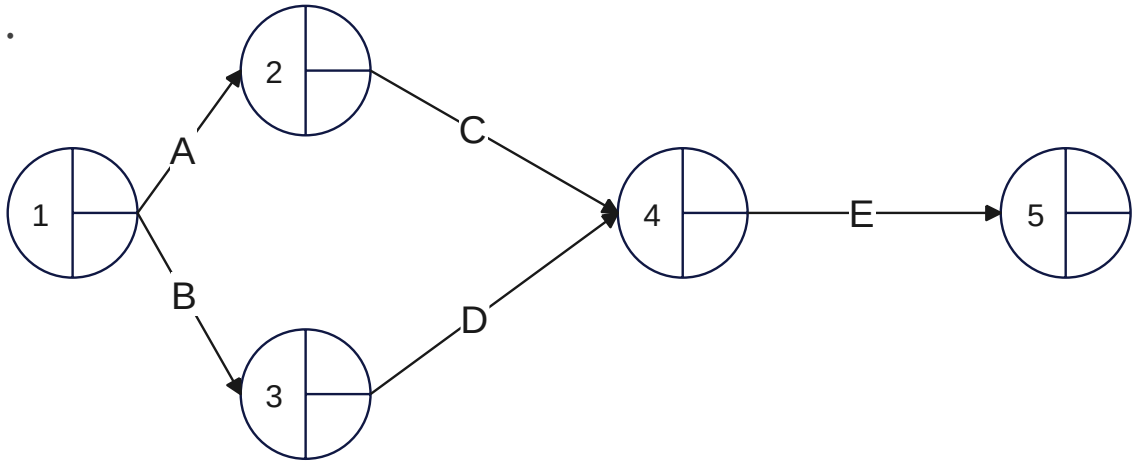
EDRAWMAX

CHARTER 1

Question 1

Activities	Dependencies
A	
B	
C	A
D	B
E	C,D

PERT Chart

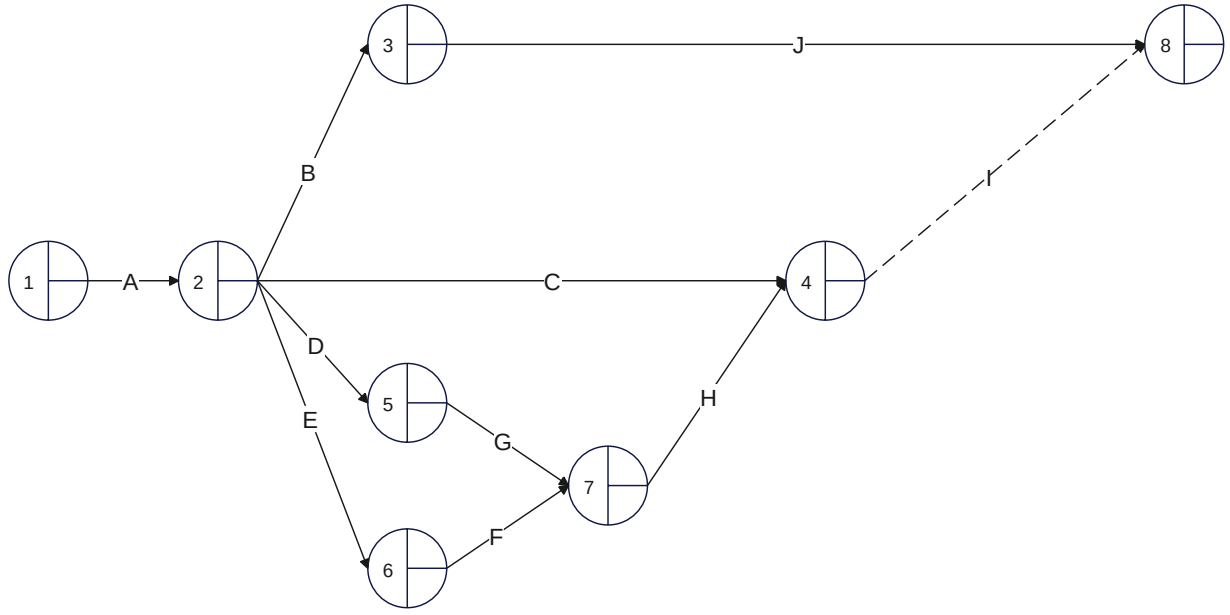


CHARTER 1 (Continues...)

Question 2

Activities	Dependencies
A	
B	A
C	A
D	A
E	A
F	E
G	D
H	G,F
I	C,H
J	B

PERT Chart



Forward & Backward Pass

SLACK, CRITICAL PATH & PROJECT DURATION

CHARPTER 2

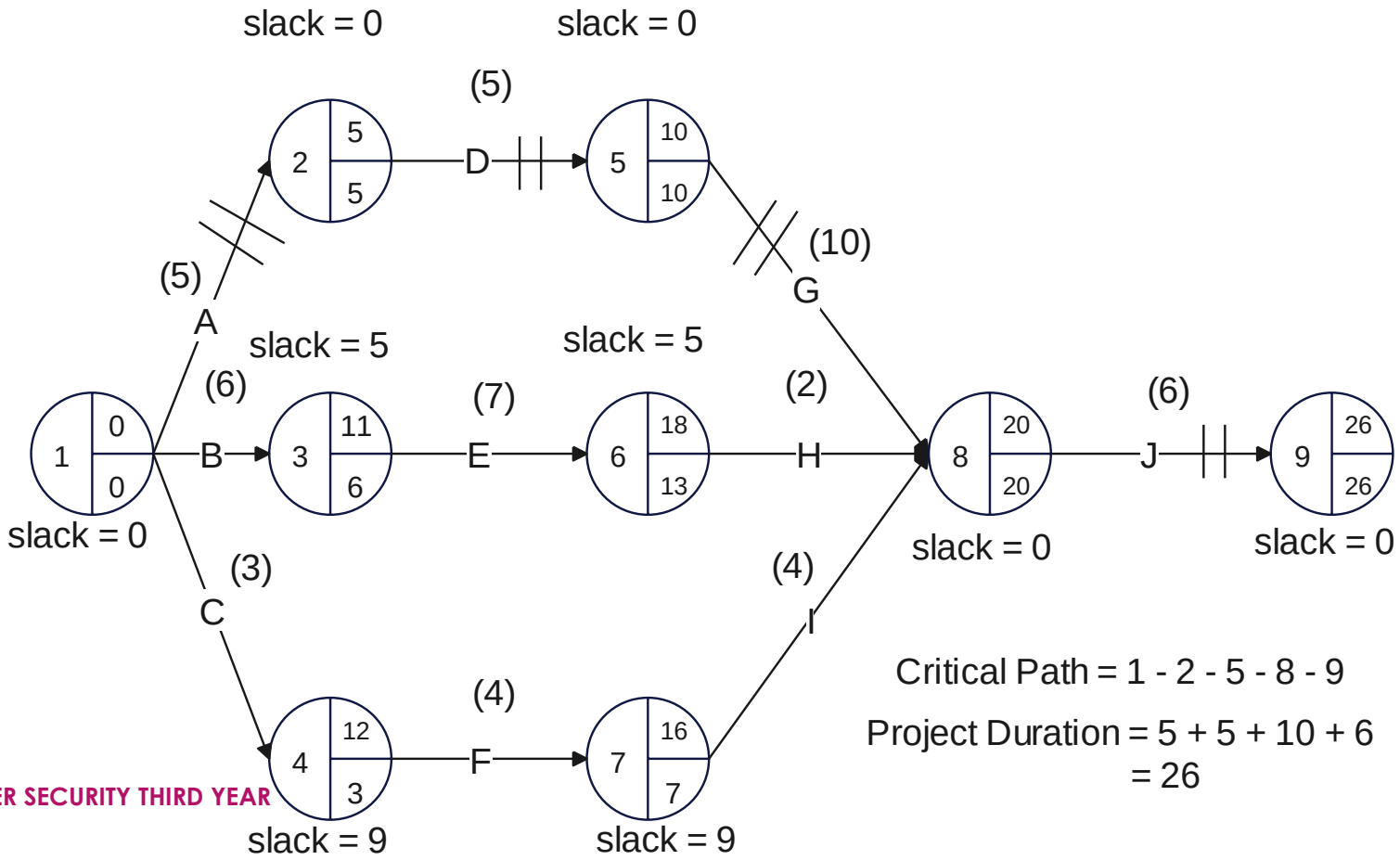
Question 1

1. Construct a network diagram.
2. Find the critical path.
3. Determine the duration the project is likely to take you to complete.

Data

Activities	Dependencies	Duration
A		5
B		6
C		3
D	A	5
E	B	7
F	C	4
G	D	10
H	E	2
I	F	4
J	G,H,I	6

CHARPTER 2 (Continues…)



Critical Path = 1 - 2 - 5 - 8 - 9

Project Duration = 5 + 5 + 10 + 6
= 26

PERT Chart

DRAWING A PERT CHART USING A STATEMENT

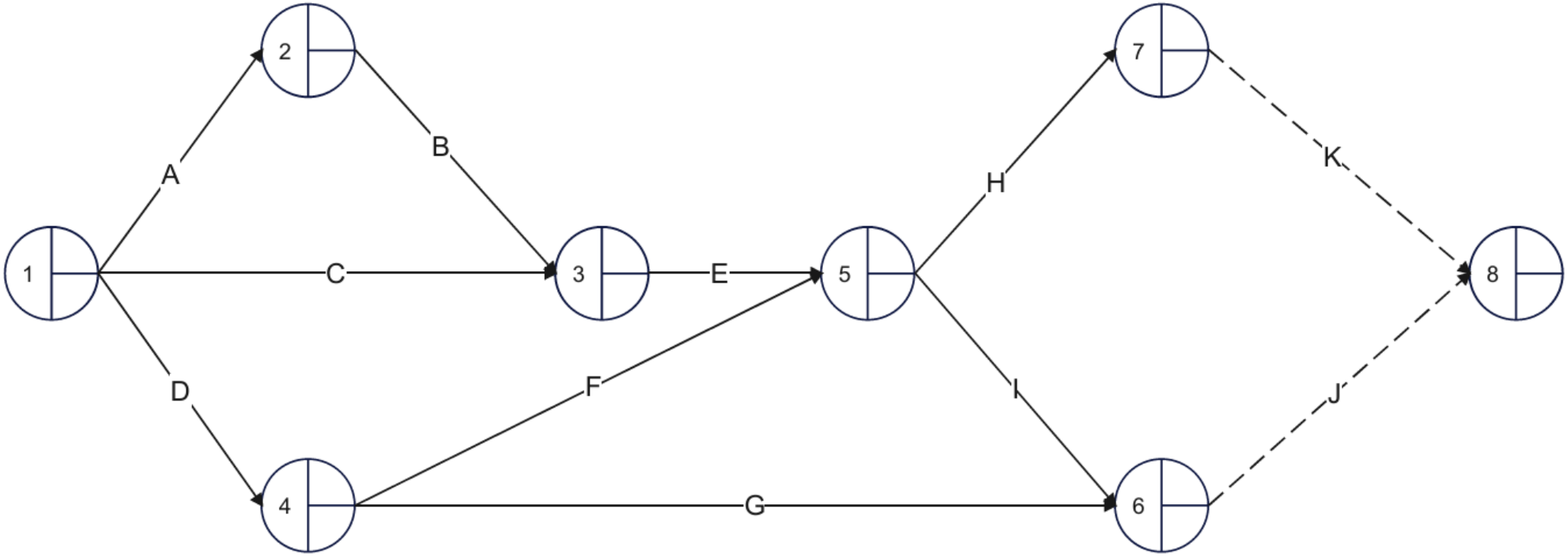
CHARPTER 3

Question 1

1. E > B, C; F, G > D; H,I > E, F; J > I, G; K > H; B > A

Activities	Dependencies
A	
B	A
C	
D	
E	B,C
F	D
G	D
H	E,F
I	E,F
J	I,G
K	H

CHARPTER 3 (Continues…)



PERT Chart

ES, EF, LF, LS, TF, FF, IND F, INT F & TAIL AND HEAD SLACK

CHAPTER 4

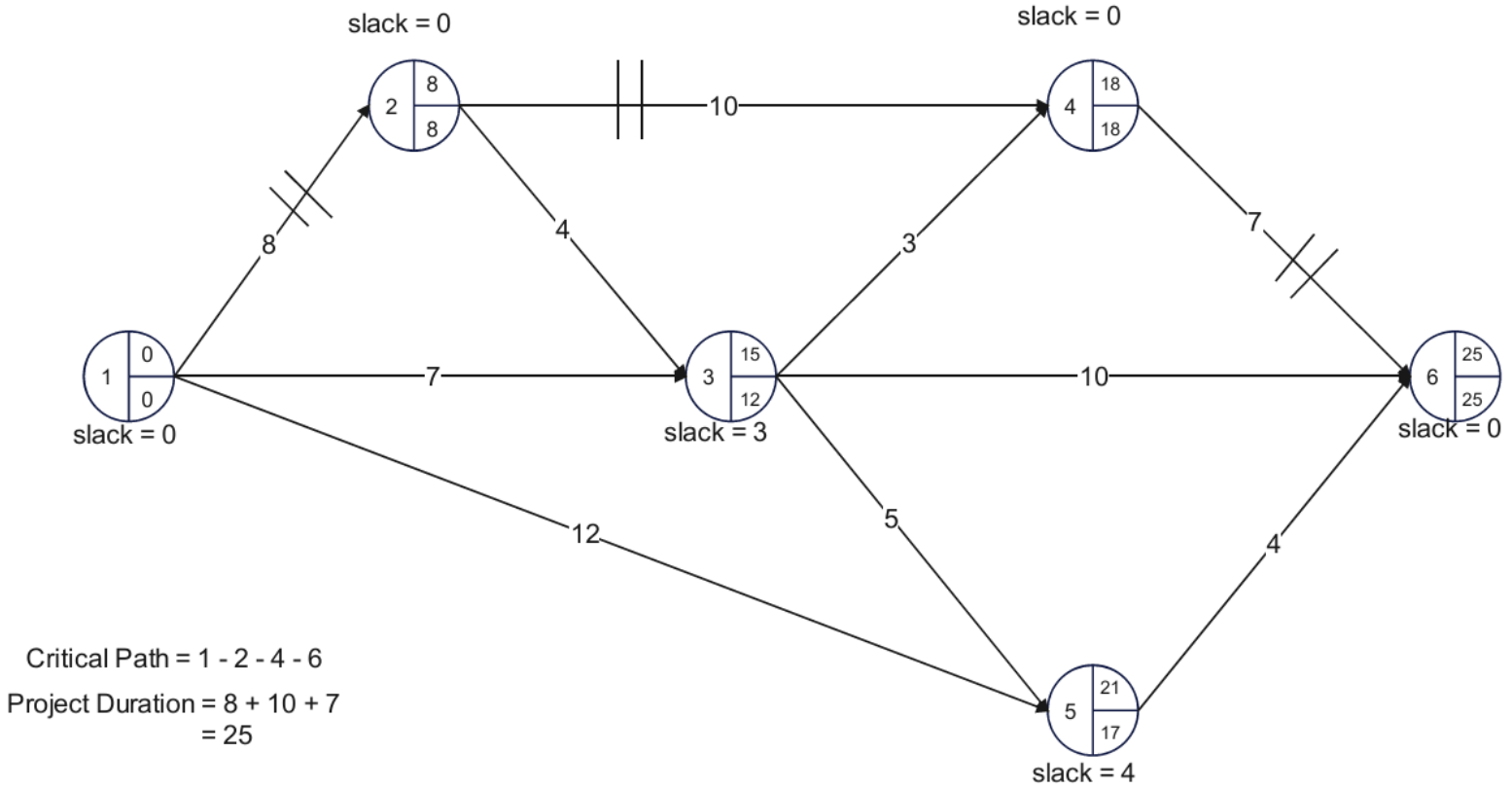
Data

Activities	Duration
1 – 2	8
1 – 3	7
1 – 5	12
2 – 3	4
2 – 4	10
3 – 4	3
3 – 5	5
3 – 6	10
4 – 6	7
5 – 6	4

Question 1

1. Calculate the Earliest Finish (EF), Eeliest Start (ES), Latest Finish (LF) and Latest Start (LS).
2. Compute the Total Float (TF), Free Float (FF), Independent Float (IND F) and Interfering Float (INT F).

CHAPTER 4 (Continues…)



Critical Path = 1 - 2 - 4 - 6

Project Duration = 8 + 10 + 7
= 25

CHAPTER 4 (Continues…)

Activities	ES	Duration (t)	EF = ES + t
1 – 2	0	8	8
1 – 3	0	7	7
1 – 5	0	12	12
2 – 3	8	4	12
2 – 4	8	10	18
3 – 4	12	3	15
3 – 5	12	5	17
3 – 6	12	10	22
4 – 6	18	7	25
5 – 6	17	4	21

CHAPTER 4 (Continues…)

Activities	LF	Duration (t)	LS = LF - t
1 – 2	8	8	0
1 – 3	15	7	8
1 – 5	21	12	9
2 – 3	15	4	11
2 – 4	18	10	8
3 – 4	18	3	15
3 – 5	21	5	16
3 – 6	25	10	15
4 – 6	25	7	18
5 – 6	25	4	21

CHAPTER 4 (Continues…)

Activities	LS	ES	TF = LS – ES
1 – 2	0	0	0
1 – 3	8	0	8
1 – 5	9	0	9
2 – 3	11	8	3
2 – 4	8	8	0
3 – 4	15	12	3
3 – 5	16	12	4
3 – 6	15	12	3
4 – 6	18	18	0
5 – 6	21	17	4

CHAPTER 4 (Continues…)

Activities	Tail Slack (T)	Head Slack (H)	TF	FF = TF – H	IND = FF – T	IND F = TF - FF
1 – 2	0	0	0	0	0	0
1 – 3	0	3	8	5	5	3
1 – 5	0	4	9	5	5	4
2 – 3	0	3	3	0	0	3
2 – 4	0	0	0	0	0	0
3 – 4	3	0	3	3	0	0
3 – 5	3	4	4	0	-3	4
3 – 6	3	0	3	3	0	0
4 – 6	0	0	0	0	0	0
5 – 6	4	0	4	4	0	0

PERT Chart

TO, TM, TP, TE, VARIANCE & STANDARD DEVIATION

CHAPTER 5

Data

Activities	to	tp	tm
1 – 2	1	3	2
1 – 3	5	7	6
1 – 4	3	7	5
2 – 5	5	9	7
3 – 5	2	6	4
5 – 6	4	6	5
4 – 7	4	8	6
6 – 7	2	4	3

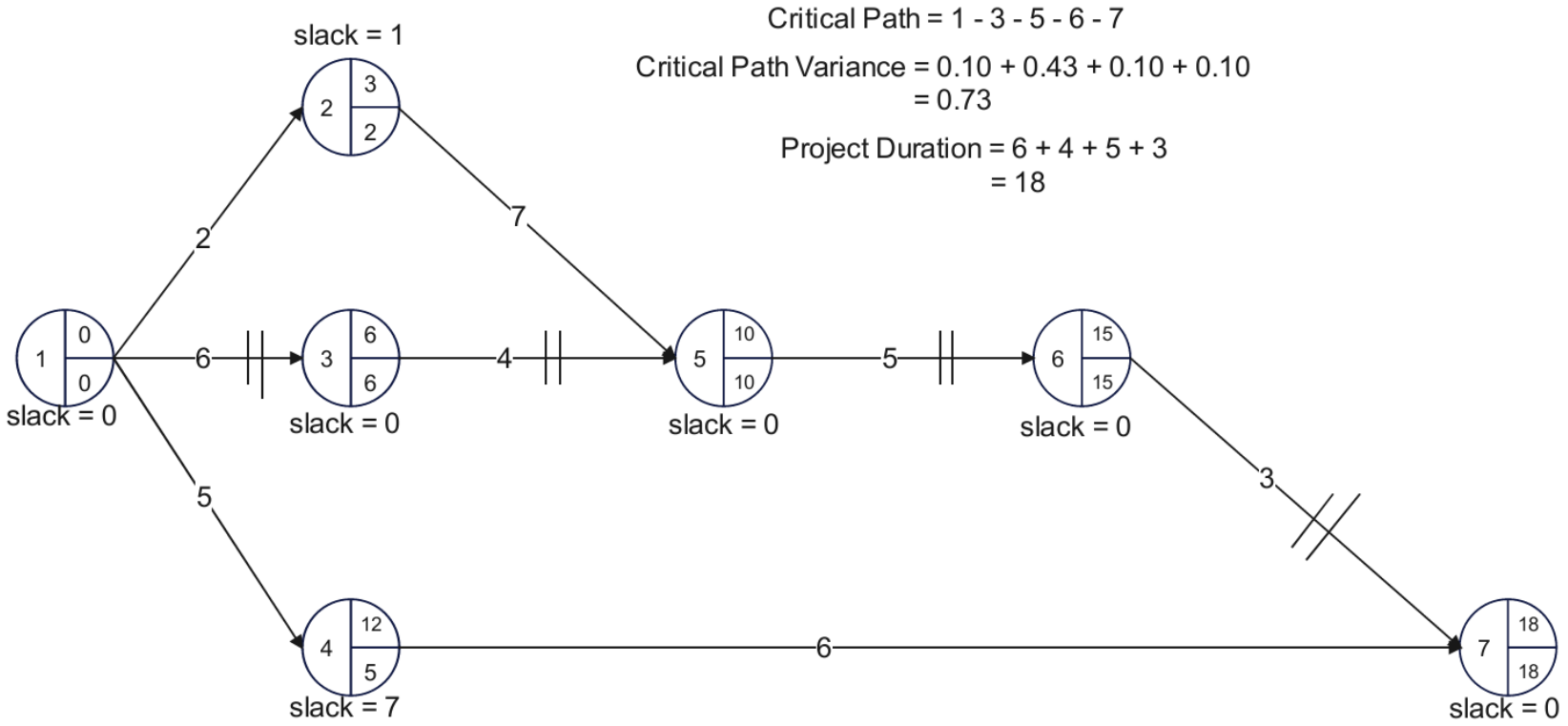
Question 1

1. Draw the corresponding network diagram.
2. Compute the expected duration for each activity.
3. Calculate the expected variance of each activity.
4. Determine the expected critical path variance.
5. Compute the standard deviation of each activity.

CHAPTER 5 (Continues…)

Activities	to	tp	tm	4tm	Te = (to + tp + 4tm) ÷ 6	σ = (tp – to) ÷ 6	σ^2
1 – 2	1	3	2	4(2) = 8	(1 + 3 + 8) ÷ 6 = 2	(3 – 1) ÷ 6 = 0.33	0.10
1 – 3	5	7	6	4(6) = 24	(5 + 7 + 24) ÷ 6 = 6	(7 – 5) ÷ 6 = 0.33	0.10
1 – 4	3	7	5	4(5) = 20	(3 + 7 + 20) ÷ 6 = 5	(7 – 3) ÷ 6 = 0.66	0.43
2 – 5	5	9	7	4(7) = 28	(5 + 9 + 28) ÷ 6 = 7	(9 – 5) ÷ 6 = 0.66	0.43
3 – 5	2	6	4	4(4) = 16	(2 + 6 + 16) ÷ 6 = 4	(6 – 2) ÷ 6 = 0.66	0.43
5 – 6	4	6	5	4(5) = 20	(4 + 6 + 20) ÷ 6 = 5	(6 – 4) ÷ 6 = 0.33	0.10
4 – 7	4	8	6	4(6) = 24	(4 + 8 + 24) ÷ 6 = 6	(8 – 4) ÷ 6 = 0.66	0.43
6 – 7	2	4	3	4(3) = 12	(2 + 4 + 12) ÷ 6 = 3	(4 – 2) ÷ 6 = 0.33	0.10

CHAPTER 5 (Continues…)



PERT Charts

REAL WORLD APPLICATION IN SYSTEM VULNERABILITY ASSESSMENT

CHAPTER 6

- This chapter will be covered on Monday, to finish our PERT Chart revisions, the venue is still M202 and time will be communicated through a WhatsApp voting poll.